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Interpretability, personalization and reliability of a machine learning based clinical decision support system

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Abstract

Artificial intelligence (AI) has achieved notable performances in many fields and its research impact in healthcare has been unquestionable. Nevertheless, the deployment of such computational models in clinical practice is still limited. Some of the major issues recognized as barriers to a successful real-world machine learning applications include lack of: transparency; reliability and personalization. Actually, these aspects are decisive not only for patient safety, but also to assure the confidence of professionals. Explainable AI aims at to achieve solutions for artificial intelligence transparency and reliability concerns, with the capacity to better understand and trust a model, providing the ability to justify its outcomes, thus effectively assisting clinicians in

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rationalizing the model prediction. This work proposes an innovative machine learning based approach, implementing a hybrid scheme, able to combine in a systematic way knowledge-driven and data-driven techniques. In a first step a global set of interpretable rules is generated, founded on clinical evidence. Then, in a second phase, a machine learning model is trained to select, from the global set of rules, the subset that is more appropriate for a given patient, according to his particular characteristics. This approach addresses simultaneously three of the central requirements of explainable AI—interpretability, personalization, and reliability—without impairing the accuracy of the model's prediction. The scheme was validated with a real dataset provided by two Portuguese Hospitals, the Santa Cruz Hospital, Lisbon, and the Santo André Hospital, Leiria, comprising a total of N = 1111 patients that suffered an acute coronary syndrome event, where the 30 days mortality was assessed. When compared with standard black-box structures (e.g. feedforward neural network) the proposed scheme achieves similar performances, while ensures simultaneously clinical interpretability and personalization of the model, as well as provides a level of reliability to the estimated mortality risk.

Keywords Explainable AI · Reliability · Interpretability · Personalization · Clinical decision support systems

1 Introduction

We are witnessing a fast dissemination of new concepts and innovative techniques in our daily life, such as artificial intelligence, self-driving cars and intelligent decision support systems. Recent research progresses on machine learning (ML) models have demonstrated remarkable performances in many different tasks and fields, from pattern recognition to natural language processing, particularly with the introduction of recent deep ML techniques (Stanford University 2016). However, despite the potential of such computational models, some barriers to fully adoption in practice remain, which are particularly evident in critical tasks, namely in clinical decision-making. In effect, real-world application of ML models has been limited due to multiple issues, including lack of trust and insufficient transparency. These aspects are decisive not only for patient safety, but also for guarantee the confidence of professionals (Qayyum et al. 2020). In effect, the black-box nature of these models often lack transparency, not being able to explain their outcomes, neither to provide a confidence level on the estimated outcomes, which is decisive in clinical field, where failure is not an option. There is also increasing societal pressures and legislation requiring that AI predictions can be explained to users, such as the European Union General Data Protection Regulation, who recently approved the "right to explanation" for any organization that uses ML algorithms to make decisions with significantly user impact (Parliament and Council of the European Union 2016). Therefore, currently there is a high motivation for innovative ML models, that can explain their reasoning, their decisions and predictions and, ultimately, to provide reliability measures in the estimated predictions.

Explainable Artificial Intelligence (XAI) is a research field, aiming at the development of techniques capable to produce more explainable models, whilst maintaining high performance (Adadi and Berrada 2018). Explanations help to provide transparency, enable assessment of accountability, demonstrate fairness in the decisions, facilitate understanding, and introduce trust on the outcomes (Arrieta 2019). In clinical domain, a higher interpretability means easier understanding and explanation of future predictions, providing healthcare experts with personalized decisions that can ultimately lead to higher quality of service in healthcare (Ahmad et al. 2018). Moreover, audits of ML models in domains like healthcare have revealed that the decisions and recommendations of such systems may be biased (Burrell 2016). Thus, interpretability is needed to ensure that such systems are free from bias, as well as they are fair in assessing different populations (Hajian et al. 2016).

Although numerous distinct approaches have been proposed, explainable AI methods can be classified based on three main principles: complexity, scoop, and level of dependency (Carvalho et al. 2019). The complexity refers to the difficulty inherent to the model, i.e., the more complex the model, the harder is to explain it. Two main approaches have been proposed to address this issue: ante-hoc and post-hoc approaches. Ante-hoc methods aim at full transparency and are typically model-specific. The simplest design translates evidence-based medical decision to computational models, e.g., decision trees. In contrast to model-based interpretability, post-hoc interpretation methods take a black-box trained model as input, and extract information about what relationships the model has learned. These approaches are typically model-agnostic, designed to work with any type of model. Considering the scoop, two main groups of techniques can be distinguished: the first focuses on personalized interpretations (local interpretability) while the second achieves explanations at a population level (global interpretability). Global explanations help to understand and to evaluate the system as a whole; local explanations help to recognize contradictions as well as to calibrate trust, on a case-by-case basis. Finally, the level of dependency refers to the scope of model application: specific or agnostic interpretability. Model specific interpretability is limited to a specific model, such as a neural network. Model agnostic interpretability is not related to a specific model, providing explanations of predictions made by any ML model.

Research on explainable AI typically faces a trade-off between ML interpretability and performance (Tjoa and Guan 2015). On the one hand, some ML models are able to provide notable interpretability results (e.g. decision trees) and, on the other hand, some ML models are capable to ensure remarkable performance results (e.g. deep learning models). In the last few years, some works have investigated and proposed new ML models, able to both guarantee interpretability and high performance, such as GA2M (Caruana et al. 2015), that investigated the performance of generalized additive models, combining single-feature models through a linear function; rule-based models like SLIM (Ustun and Rudin 2016); and Bayesian rule approaches (Wang et al. 2017). ANCHORS is an example of the current state of-the-art high performance explainable rule-based model (Ribeiro et al. 2018). It is a model-agnostic system, able to explain predictions generated by a generic ML model with high precision. The model provides rules, designated as anchors, to explain the prediction for each occurrence. However, the practical utility of these models in healthcare has not been convincingly demonstrated, mainly due to the difficulty to understand the involved concepts from the clinical perspective. Thus, to be effectively adopted by clinical community, ML schemes should be able to directly incorporate clinical evidences, guaranteeing models' interpretability, ensuring the trust of professionals and effectively contributing to the reliability of the decision making.

In order reach this objective, this work proposes an hybrid approach towards the development of more trustable and explainable ML based decision support systems, through the combination of clinical evidence with ML models. The ultimate goal of this approach is to contribute to ML based models that can effectively be adopted within the clinical workflow, improving the quality of the clinical decision. The proposed approach addresses interpretability, personalization and reliability issues in a systematic way, without impairing the performance of standard ML strategies. It comprises three main phases. The first phase involves the creation of a set of rules directly based on the clinical patient's characteristics (isolated or combined), able to describe in an interpretable form for clinical reasoning, thus defeating the black-box limitations of common ML models. The second phase involves the implementation of a personalization mechanism, through the selection of the subset of rules that are more appropriate for a specific patient, thus avoiding the principle "one fits all", typical of common ML models. The third phase implements an ensemble strategy, using the outcomes of the selected rules to compute a global prediction and, simultaneously, provides a reliability measure for this estimation. In this way, a degree of confidence about each individual outcome is provided to the physicians.

The validation of this broad methodology is focused on a decision support system for the mortality risk assessment, using two Portuguese real dataset provided by the Santa Cruz hospital (Lisbon) and Santo André Hospital (Leiria). The global dataset comprises $N = 1111$ patients that suffered an acute coronary syndrome event, presenting a 30 days mortality rate of 4.95%. The paper is organized as follows: Sect. 2 overviews the related work, Sect. 3 describes the proposed methodology, Sect. 4 presents and discuss the results, being the conclusions detailed in Sect. 5.

2 Related work

This work focus the development of machine learning schemes, aiming to overcome some of the basic requirements of explainable AI, while not impairing the accuracy of the model's predictive performance.

2.1 Main concepts

2.1.1 Interpretability versus explainability

In the context of explainable AI, there are two fundamental concepts: "interpretability" and "explainability". Although there are no rigorous and clear definitions about these concepts, interpretability is usually associated with mathematical notions, reflecting the mathematical operation of the model. Examples are the inherently interpretable models, characterized by structures and parameters to which a certain interpretation can be associated (Rudin et al. 2013). In this context, linear and logistic regression models, decision trees and decision rules are considered to be interpretable (Freitas 2014). On the other hand, explainability concentrates on understanding why a model produces a particular result, without addressing all the mathematical aspects. Thus, according to these definitions, while interpretability is based on reliable mathematical rules, explainability can be unfair, biased, or meaningless to humans, since it may not follow the expected real-world rules for a specific domain (Selvaraju et al. 2019). As result, the interpretability can be seen as a prerequisite for the explainability and, additionally, a set of interpretable judgments is only considered explainable in the particular context of the domain knowledge where it is applicable.

2.1.2 Explainable AI requirements

The predictive performance (accuracy) of recent ML methods has achieved notable results. However, predictive performance is not the only significant aspect when developing a model. In effect, for critical and/or sensitive domains, a detailed understanding of the model can be more relevant than its accuracy. This understanding is fundamental to overcome some of the barriers that have constrained the fully adoption of ML models, particularly evident in critical tasks, namely in clinical decision-making. To achieve this goal, interpretability and explainability are essential concepts to fulfil some of the essential requirements to overcome such barriers. Among these requirements we can mention causability, ethical decision-making issues, accountability, fairness and trust (Burkart and Huber 2021).

Causability: The concept of causability, introduced in Holzinger et al. (2019), is of major importance in certain domains, namely in the clinical context. It should be noticed that causability is not a synonym for causality in the traditional sense (the relationship between cause and effect). In effect, whilst explainability concerns to the understanding and transparency aspects of ML approaches, causability is about measuring and ensuring the quality of explanations (Holzinger et al. 2021). Therefore, causability measure extents human user explanations to a level of realistic causal understanding, quantified with effectiveness, efficiency and satisfaction in a specified context of use, similar to the usability concept. In this context the same authors (Holzinger et al. 2020) have proposed the System Causability Scale, a simple and efficient assessment tool capable to measure the quality of an explanation interface (human–AI interface) or the explanation process in a given model.

Ethical decision making: The requirements related with ethical, legal and regulatory challenges are decisive in the health area, since clinical decisions usually have a direct impact on the condition of a patient. Addressing these principles, the European Commission has recently stated (European Commission 2020) that "… while a number of the requirements are already reflected in existing legal or regulatory regimes, those regarding transparency, traceability and human oversight are not specifically covered under current legislation in many economic sectors". Thus, the future of ML anticipated by the EU will require the introduction of the "human in the loop as a necessary safeguard" to the final decision, not allowing decisions exclusively based on automated processing (Schneeberger et al. 2020). Moreover, ML predictions have to explain to users their results in a comprehensible way, guaranteeing the "right to explanation" requirement, essentially to observe conformity to ethical standards and to the principles of accountability and retracement, while contributing to introduce reliability and confidence in the predictions (O'Sullivan 2019).

Fairness: In parallel, explainable AI has established fairness as an important issue. Actually, ML modes can face fairness issues, when they are applied to specific populations (Nassih and Berrado 2020), particularly in sensitive areas, such as health. Therefore, it is critical to ensure that decisions are made in a total fair and transparent way, not showing discriminatory conclusions for specific groups (Barocas et al. 2001). In (Mehrabi et al. 1908), the authors analyse different real-world applications, enabling to identify different sources of biases that can affect ML applications. Additionally, they propose a taxonomy for the fairness, aiming at avoiding bias in ML models.

Trust: Trust is one of the major requirements for explainable AI, in particular in clinical domain. Recognize and understanding the model's strengths and weaknesses is a central prerequisite for human trust and, hence, for model confidence and acceptance (Markus et al. 2020). This topic, i.e. the assessment of model's trust has been researched by many authors, usually associated with the interpretability, as an approach to provide a transparent reasoning on a model operation. In (Caruana et al. 2015), a generalized additive model with pairwise interaction (GA2Ms) is proposed, enabling to achieve a trade-off between model accuracy and intelligibility, facilitating the deployment of high-accuracy models. The solution was applied to decide which pneumonia patients arriving at the hospital should be admitted and which patients should be permitted to go home for outpatient care.

2.2 Interpretability and explainability assessment

2.2.1 Interpretable models taxonomies

The development of techniques capable to generate interpretable models has been an active research topic, where numerous distinct approaches have been proposed. In order to systematize the different interpretable models, several taxonomies have been proposed (Burkart and Huber 2021; Linardatos et al. 2021). Basically, interpretable models are categorized in two general categories: intrinsically interpretable models (ante-hoc) and post-hoc explainability methods.

Intrinsically interpretable models: in this case the main principle is the use of algorithms to directly generate interpretable models. Among these, two main group of ML models can be recognized: tree-based algorithms such as CART (Breiman et al. 1984), ID3 (Quinlan 1986), C4.5 (Quinlan 1992), and the rule-based algorithms such as RIPPER (Cohen 1995), M5 Rules (Holmes et al. 1999), and more recently RIPE (Hastie et al. 2001) and Covering Algorithm (Hornik et al. 2009). The explanations are direct and, as consequence, decisions are easily understandable. Additionally, model's structure must be kept simple to be easily explained since, for example, a decision tree composed of 50 rules will not be easily understandable by humans (Molnar 1990). However, intrinsically interpretable models generally faces some accuracy limitations when applied to complex tasks, as they form a restricted category of models, when compared to more complex machine learning models (e.g. neural networks and deep learning models).

Post-hoc methods: Post-hoc methods addresses the limitations of interpretable models by decoupling the model from its explanations, introducing flexibility in model interpretability, since the explanations do not depend on the specific ML model. Usually, two main approaches are followed for the generation of interpretable post-hoc models. The first is based on the explanation of the importance of each individual feature (inputs) with respect to the prediction outcome; the second relies on the creation of an interpretable model (surrogate model) founded on the ML model previously obtained (Burkart and Huber 2021; Linardatos et al. 2021).

Explaining individual features: Typically these methods are model-agnostic (able to work with any type of model), and are based on the study of the sensitivity of the ML model, by manipulating input data and analysing the respective model outcome. Two classes of methods can be distinguish: local and global. Local methods, explaining individual predictions of ML models, includes Shapley Values (Strumbelj and Kononenko 2014) and Counterfactual Explanations (Dandl et al. 2004; Mothilal et al. 2020) methods. Counterfactual explanations explain predictions in the form of what-if scenarios. The Shapley values, originated from collaborative game theory, provide an answer on how to fairly share a pay out among the players of a collaborative game. Another example is Saliency Maps (Adebayo et al. 2018), an interpretation method specific for CNNs, that makes use of the network gradients to explain individual classification. The explanations are in the form of heat maps that show how changing a pixel can change the classification. Global model-agnostic explanation methods are used to explain how the model behaves on average for a given dataset. Permutation Feature Importance (Casalicchio et al. 2018) and Conditional Variable Importance methods (Strobl et al. 2008), are instances of such methods, based on the relative importance assessment of each feature for the output prediction.

Surrogate Models: Surrogate models are interpretable models developed to reproduce the behaviour of a general ML model. Thus, although the ML model is assumed to be black-box, its inputs and outputs are used to train a surrogate ML model. As result, the interpretation is possible, based on the explanation of the interpretable surrogate model. Several methods have been proposed in this context (Puri et al. 2017; Frosst and Hinton 2017; Krishnan and Wu 2017). LIME (Ribeiro et al. 2016) is a well-known example of a local surrogate method that explains individual predictions by learning an interpretable model with data in proximity to the operating point to be explained.

2.2.2 Interpretability measures

Being the interpretability, by definition, an imprecise concept, the definition of a rigorous measure to quantify the level of a model's "interpretability" is a challenging task to explainable AI (Lipton 2018). The quantification of the predictive performance of an ML model can be easily obtained through the prediction error, i.e., the comparison between the ML outcomes and the ground truth labels. However, the quantification of the interpretability of the same ML model is much more challenging. In effect, the ground truth explanations are not known and, therefore, it is not so straightforward to quantify the interpretability or the truthfulness of an explanation (Zhou et al. 2018).

As result, instead of trying to directly quantify the interpretability, several aspects connected with interpretability have been assessed, including sparsity, interaction strength, stability, simplicity and complexity aspects (Molnar et al. 2019) (Philipp et al. 2018). In this context, in Margot (2020) it is proposed an accurate mathematical definition to the interpretability concept, based on three individual metrics easily quantified by simple formulas: predictive performance, stability and simplicity. In the same perspective (Murdoch et al. 2019) introduced the PDR framework, consisting of three distinct metrics to assess the interpretability of a model: predictive accuracy (P), descriptive accuracy (D), and relevancy (R). The first two are automatically computed, the last involves the assessment of a human user.

In particular, for the predictive performance (accuracy) assessment, several metrics have been proposed, namely to deal with imbalanced data (Dubey et al. 2014), which is very common in clinical contexts. Examples of such metrics are the geometric mean and the F1 score. These measures are single operating point metrics, i.e., defined to a specific threshold, unable to provide a global performance at all operating points. A typical alternative to a global operating measure, particularly useful in the clinical domain, is the area under the receiver operating characteristic curve (AUC) metric (Bradley 1997). However, this measure is too general, including operating points that are not used in practice and does not provide information about the distribution of the performance along the curve. In (Carrington et al. 2021) an in between method was proposed, the deep ROC analysis. This method is especially relevant in the context of explainable AI, since it can provide interpretation and assurance for patients in each risk group, therefore improving the assessment of model's performance.

2.2.3 Explainability assessment

Two main factors determine the explainability of an ML model: the parameters of the model and the human's capacity to understand it (Zhou et al. 2021). Founded in this principle, several approaches have been carried out in order to categorise the existing explainability methods (Guidotti et al. 2018). In particular a taxonomy based on three distinct categories was proposed to assess the explanation of a ML model (Doshi-Velez and Towards 2017), consisting of the application-grounded evaluation, the human-grounded evaluation and the functionality-grounded evaluation. The first (experiments with end-users) addresses the specific effect of the interpretation process in the end-user (human with domain knowledge), for example, how well explanations can assist the end-user in the decision making. The second (experiments with lay humans), although similar to the first one, is more general and does not require that human is a domain expert. Finally, the last does not require experiments involving the human, being the evaluation of the explanation based on some formal definition of interpretability (e.g., the depth of a decision tree). The first two categories are human centred evaluations, therefore, although being subjective, they can provide a strong evidence of the success of explanations. The last category provides an objective quantification.

2.3 Proposed approach

Explainable AI is facing a debate around the use of intrinsically interpretable models and/or post-hoc methods. On one hand, instead of creating models that are inherently interpretable, there has been a recent explosion of research on post-hoc techniques to explain black box models, particularly in the deep learning context. However, the obtained explanations might be unreliable and, eventually, misleading. Therefore, there is some scepticism regarding the use of these models in scenarios which may require critical decision making, such as in the clinical domain. On the other hand, the use on inherently interpretable models, although appealing, typically achieves low predictive performances, when compared with black-box models. In effect, the use of a limited number of rules applied indistinctively to heterogeneous groups of individuals (the general population), could result into global lower predictive performance results.

The scheme proposed here can be included in the inherently interpretable models approaches. However, our main assumption is that the initial set of interpretable rules should not be applied in the same way to all individuals. Actually, simple explanations for a specific individual have to be provided, using a subset of global model's rules. To this aim, we propose an hybrid scheme, combining knowledge-driven methods (to create the set of rules for the general population) with data-driven methods (to select the most appropriate subset of rules for each individual). As result, although ML methods are used in this scheme for data-driven purposes, they are employed from a different perspective, when compared with the common approach. Instead of training a ML model to derive the set of rules (common approach) to predict the target output, the ML models are trained to select the subset of rules that has more potential to fits a specific individual, thus providing personalized explanations. Therefore, the interpretability, explainability as well other requirements such as accountability, causability and trust are directly assured. In the present work, the baseline set of rules, in the form of "*if then*" rules, is created by means of a two-step procedure. In the first step the thresholds of the rules are computed through a data-driven optimization process (maximizing the accuracy of each rule) being, in a second phase, these thresholds adjusted by the clinical team, based on its expertise.

Additionally, in order to increase the trust of a model, besides the understanding of its behaviour, clinicians should have confidence on the predicted outcomes. Therefore, this work performs a step forward in that direction, proposing a measure of confidence, able to complement the predicted outcome of the ML model with an estimation for its reliability.

Finally, the strategy is evaluated using two complementary metrics: the geometric mean (GM) and the area under the receiver operating characteristic curve (AUC), respectively a single operating point metric and a global metric.

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3 Methodology

3.1 Notation

In this work we consider a data-driven supervised context where, given a m-dimensional feature vector $\boldsymbol{x} \epsilon R^{1,M}$, the goal is to estimate a target $t \epsilon R$. To this aim we assume the training of a machine learning model capable to map the vector $\boldsymbol{x}$ (input) into the scalar t (output), $NN : \boldsymbol{x} \rightarrow t$. Moreover, we consider the training data for the machine learning model to be composed by the N tuples $\{\boldsymbol{x}_i, t_i\}_{i=1}^{N}$ and we denote the specific j^{th} feature from i^{th} instance as $x_{ij} i = 1..N,\ j = 1..M$. As result, the data set can be defined by an input matrix and an output vector, respectively, $\boldsymbol{X} \epsilon R^{N,M}$ and $\boldsymbol{t} \epsilon R^{N,1}$.

In particular, the objective of this specific classification problem is the discrimination of cardiovascular patients into one of two possible mortality risk classes, thus $t = \{0{,}1\} = \{\text{survival, death}\}$. To this purpose the features, $\boldsymbol{x} \epsilon R^{1,M}$, consists of a set of risk factors (inputs), from distinct data types. Examples of such risk factors are age: AG $\in \{30,\ldots,110\}$, gender: GD $= \{0{,}1\} = \{$male, female$\}$ and number of comorbidities: NC $\in \{1{,}2, \geq 3\}$, respectively of, continuous, binary, and categorical data type.

For sake of clarity, namely to allow a direct comparison, two different machine learning approaches are described: (1) a standard machine learning based approach; (2) our proposed methodology that relies on an innovative machine learning based method. It should be emphasized that this new methodology can be applied to a broad set of classification problems. However, for explanation issues and without loss of generalization, it is described based on the particular context of cardiovascular risk assessment. Moreover, although a neural network classifier is used as ML model, other ML structures could be used without loss of generality.

3.2 Standard machine learning based approach

Using a standard ML approach, e.g. a neural network (NN), the model would be trained employing a data-driven supervised method, based on an available training dataset. The inputs are the risk factors of the patients,$\boldsymbol{X} \epsilon R^{N,M}$, and the output is the actual mortality outcome $\boldsymbol{t} \epsilon R^{N,1}$, as depicted in Fig. 1, being M and N, respectively, the number of risk factors and the number of instances (patients). Example of a specific

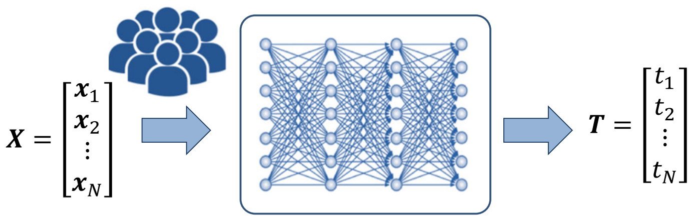

Fig. 1 Training of a standard machine learning model: inputs and target

input is $\boldsymbol{x} \epsilon R^{1,3}$=$\{AG, GD, NC\} = \{78, 0, 1\}$, being the target $\boldsymbol{t}$ the respective mortality class t=$\{survival\} = \{0\}$.

Given the universal approximation properties of a NN, a classification model able to achieve high performances, i.e., low values of error (difference between the actual output and the output estimated by the NN) can be achieved. Additionally, to increase the likelihood of the generalization properties (i.e., the capacity of the model to be well succeed when applied to unobserved data) the splitting the data in training and testing subsets has to be carried out. These validation techniques apply, among others, holdout, regularization and cross validation mechanisms (Ying 2019).

After being trained/tested using the initial training/testing dataset, the performance of the NN model can be afterwards assessed in real operation. To this aim, the particular characteristics of a new patient, $\boldsymbol{x}_i \epsilon R^{1,M} i = 1, \cdots N$, are fed to the NN model and the estimated NN output ($\widehat{t_i} \epsilon R$) is compared with the desired target ($t_i \epsilon R$), obtaining an error $e_i \epsilon R$, as depicted in Fig. 2.

In binary problems, where the output only assumes two values, the performance can be quantified using well-known metrics, such as sensitivity (SE) and specificity (SP), respectively determined by Eq. (1) and (2).

$$SE = \frac{TP}{TP + FN} \tag{1}$$

$$SP = \frac{TN}{TN + FP} \tag{2}$$

True positive (TP) and false negative (FN) represent the number of death cases ($t_i = 1$) that have been, respectively, correctly ($\widehat{t_i} = 1$) and incorrectly identified ($\widehat{t_i} = 0$). True negative (TN) and false positive (FP) represent the number of survival cases ($t_i = 0$) that have been, respectively, correctly ($\widehat{t_i} = 0$) and incorrectly identified ($\widehat{t_i} = 1$). Moreover, in clinical studies, the ability to correctly predict the output is often assessed through the area under the receiver operating characteristic curve (AUC) (Hajian-Tilaki 2013). The cardiovascular risk assessment is a highly imbalanced scenario, as there is a high number of survival cases compared to a much lower number of death cases. In such problems cases, where a class is highly prevalent over the other, the accuracy (arithmetic mean of SE and SP) is not a suitable performance measure and should be replaced by other metrics, like the geometric mean (GM) (Luque et al. 2019),

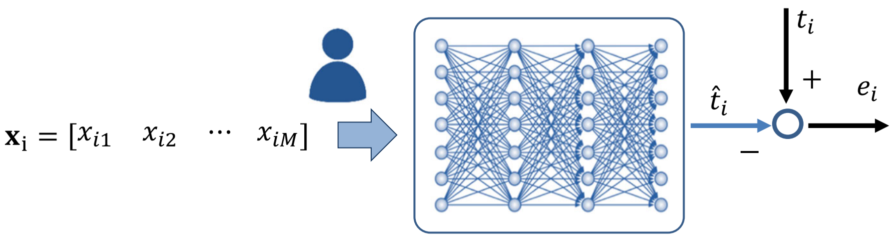

Fig. 2 Validation of a standard machine learning model for a new subject

computed from the SE and the SP as (3).

$$GM = \sqrt{SE \times SP} \tag{3}$$

Although NN models (or other machine learning structures) can reach remarkable performances, e.g., GM or AUC, they usually fail to comply with respect to some of the major requirements for the effective acceptance of decision support systems: explainability, interpretability, personalization and reliability in the obtained estimations. In effect, as mentioned, standard machine learning structures are black-box models and, thus, they are not directly explainable or interpretable. Moreover, they follow the principle one fits all (i.e. the model is applied in the same way to all patients), without the capacity to be tailored to the specific characteristics of a patient. Definitively, these flaws do not allow an effective application of these classification models into the decision process in the daily clinical practice. Moreover, new regulations have imposed the mandatory audit and verifiability of decision support systems, increasing the need for the ability to understand and trust ML models, for which interpretability is indispensable. Thus, these weaknesses imposed the development of different ML approaches.

3.3 Proposed machine learning based approach

When compared with the standard ML approach, described in 3.2., this innovative methodology uses the ML approximation capabilities from a new perspective, being composed of three distinct phases.

The first phase relies on the creation of a baseline set of interpretable rules, directly derived from the available risk factors data, that can be complemented by rules provided by clinical experts. In this way the incorporation of clinical knowledge in the current strategy is straightforward, achieved by the definition of a set of rules, that incorporate clinical know-how.

The second phase implements a ML based scheme, capable to identify the subset of rules that is more appropriated to be applied to a specific patient, i.e., select the best subset of rules for a specific patient.

The third phase uses and combines this subset of rules to estimate the actual cardiovascular risk of the patient, together with an index of its confidence/reliability. The key point of this different perspective is the paradigm of how the ML model is employed. In to some extent, this strategy simulates the reasoning of a physician: based on his clinical expertize (reflected in the baseline set of rules) and facing the specific characteristics of a patient (available risk factors), the physician selects and applies only the rules that are more appropriated to that particular diagnosis.

3.3.1 Phase 1: Derivation of baseline of rules—Interpretability issue

Rules structure definition The first step of the current approach consists in the specification of a set of explainable rules, to be used in the classification/diagnosis process. These rules must have a clinical meaning, ideally according to the current

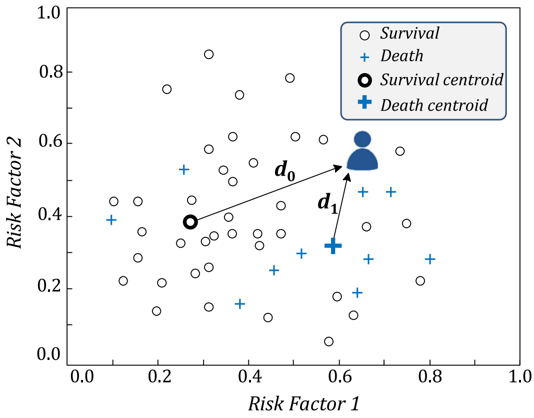

Fig. 3 Clustering process and classification based on the distance (similarity) to centroids

practice followed by physicians, thus directly interpretable and accepted by them. Any type of interpretable rules can be straightforwardly integrated in this strategy. For example, simple rules that binarize individual risk factors, in the form of (4), are suitable to tackle this issue, i.e., to estimate the mortality risk ($\widehat{t_i}$) of a particular patient.

$$if AG \geq 80 then \widehat{t_i} = 1 \tag{4}$$

Moreover, more complex rules combining two or more risk factors, can also be considered, e.g. $if(GD = male AND DNC \geq 3)$ *then* $\widehat{t_i}$=1. These rules can be created based on knowledge driven approaches (considering clinical evidence) or following a data driven approach. In the first case the rules' structure and respective thresholds can be defined using the experience of clinical professionals or, alternatively, from the clinical literature. In the second scenario, the available data is used to obtain such rules.

Rules based on "virtual patients" Without loss of generality, a data driven approach is assumed here. In particular, data-driven rules are obtained based on "virtual patients" concept, obtained through clustering strategies, assuming that patients with similar characteristics can be grouped. Following this approach, two centroids are determined from the training dataset (risk factors), each one representing a class: the values of the patients who died and the values of the patients who survived. These two centroids represent the typical characteristics of two virtual patients: the {dead} and {survival} ones. Having determined such centroids, the distance (i.e. the similarity) of a new patient to these centroids can be used to classify the new patient: if he is closer to the "death centroid", thus he is expected to die; otherwise, if he is closer to the "survival centroid", the patient is expected to survive. In fact, this reflects the reasoning of a physician about a given patient: the clinician uses his/her past experience, recalls similar cases and applies his/her historical knowledge to the specific case under diagnosis.

Figure 3 exemplifies the result of this clustering approach considering the combination of two risk factors (risk factor 1 and risk factor 2). For this illustrative example, since the distance from the patient to the "death centroid" (d_1) is smaller than the distance to the "survival centroid" (d_0), the mortality risk of patient will be classified as {death}.

In order to generalize the methodology for all the risk factors (eventually presenting centroids with quite different absolute values), the distance of each patient to the centroids is determined through a normalized distance d_n, given by (5).

$$d_n = 1 - \frac{d_1}{d_1 + d_0} \tag{5}$$

The variable d_1 represents the Euclidean distance of a patient to the "death centroid", and the distance d_0 represents the Euclidean distance of a patient to the "survival centroid" one. As result, this metric varies between $d_\mathrm{n} = 0$ (when the patient coincides with the "virtual patient" that represents the survival group) and $d_\mathrm{n} = 1$ (when the patient coincides with the "virtual patient" that represents the death group). Therefore, this normalized distance can be used to define, similarly to Eq. (4), a rule for the classification task, according to (6).

$$if d_n \geq L then \widehat{t_i} = 1 \tag{6}$$

The value of L represents the threshold that categorizes the patient into the death or survival group. In the standard case, the mean distance between the two centroids determines that binarization, thus $L = 0.5$.

Furthermore, the computation of "virtual patients" was considered independently for each risk factors. Consequently a set of rules equal to number of risk factor is obtained, being each one of the rules associated to a specific risk factor. In this scenario, such consideration has the advantage that it allows to, afterwards, converted the threshold L to absolute values, obtaining rules in the form of (4). In other words, the rule not only states to which "virtual patient" the case under diagnosis is more similar, but it is also possible to know the absolute value of L (for example, $L = 80$ for risk factor AG).

Threshold computation (L): geometric mean optimization.

Although a threshold value of $L = 0.5$ might be acceptable, it does not assure the maximum accuracy obtained by each rule. Therefore, in order to maximize the performance of each rule, a method to compute an optimal value for the threshold L is proposed.

To this aim the geometric mean (GM) is selected as the criterion to be maximized. For a binary classification task, when only two values are possible, {0,1}, the sensitivity and specificity, Eqs. (1) and (2), can be rewritten as (7) and (8), respectively.

$$SE = \frac{1}{T_1}\left(\boldsymbol{t}' \cdot \widehat{\boldsymbol{t}}\right) \tag{7}$$

$$SP = \frac{1}{T_0}\left((1-\boldsymbol{t})' \cdot (1-\widehat{\boldsymbol{t}})\right) \tag{8}$$

The operator $(\cdot)$ represents the dot product between the two vectors $\left(\boldsymbol{t}' \cdot \widehat{\boldsymbol{t}}\right)$, where $\left(\boldsymbol{t}'\right)$ represents the transpose of the target risk vector $\boldsymbol{t} \epsilon R^{N,1}$, and $\widehat{\boldsymbol{t}} \epsilon R^{N,1}$ the estimated

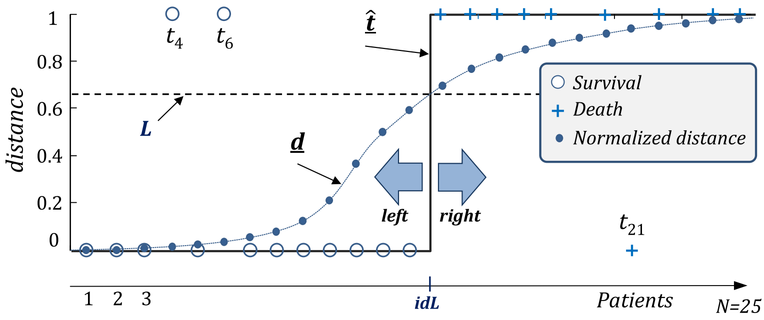

Fig. 4 Establishment of the threshold for the decision rule

risk vector using the classification rule (6). The scalars T_0 and T_1, quantify, respectively, the number of survival patients and the number of death patients existent in the target vector $\boldsymbol{t}$. In effect, since only two values are possible $\{0,1\}$, the operation $\left(\boldsymbol{t}' \cdot \widehat{\boldsymbol{t}}\right)$ quantifies the number of values that are simultaneously equal to $\{1\}$ in the vector $\boldsymbol{t}$ and in the vector $\widehat{\boldsymbol{t}}$, thus determining the sensitivity. A similar procedure applies to the operation $(1-\boldsymbol{t})' \cdot \left(1-\widehat{\boldsymbol{t}}\right)$ to quantify the specificity, i.e., the values that are simultaneously equals to $\{0\}$ in the two vectors. As result, the maximization of the GM, with respect to the threshold L, can be formulated as (9). The terms T_0 and T_1 are fixed and, therefore, can be discarded. Moreover, to avoid the squared root operator, the square of the GM was considered as the value to be maximized. It should be noted that these procedures do not affect the maximization process.

$$\max_{0 \le L \le 1} GM^2 = (t' \cdot \widehat{t})\left((1-t)' \cdot (1-\widehat{t})\right) \tag{9}$$

From the application of the classification rule to the N patients, Eq. (6), a vector of normalized distances in the interval [0,1] is obtained, $\boldsymbol{d} \epsilon R^{N,1}$. Let us assume the matrix $\boldsymbol{M} \epsilon R^{N,2}$, composed of the vectors $\boldsymbol{d}$ and $\boldsymbol{t}$, i.e., $\boldsymbol{M} = [\boldsymbol{d}\boldsymbol{t}]$. Additionally, let us consider a data transformation process, sorting the matrix $\boldsymbol{M}$ with respect to the distance $\boldsymbol{d}$ (from the lowest to highest values). As result of this transformation a ordered matrix, $\underline{\boldsymbol{M}} = \left[\underline{\boldsymbol{d}}\underline{\boldsymbol{t}}\right]$ is obtained, and the maximization procedure can be rewritten as (10) (note that only a re-order of the values is involved).

$$\underset{0 \le L \le 1}{\text{Max}}\, GM^2 = \left(\underline{\boldsymbol{t}}' \cdot \underline{\widehat{\boldsymbol{t}}}\right)\left((1-\underline{\boldsymbol{t}})' \cdot (1-\underline{\widehat{\boldsymbol{t}}})\right) \tag{10}$$

The maximization of the GM metric aims at finding the value of the threshold L, and the respective localization of index idL, that maximize (9), as depicted in Fig. 4. The index idL identifies the transition between the survival (left) and death (right) patients that are differentiated by the rule.

Actually, the maximization procedure can be interpreted as follows: the objective is to define the index idL that ensures the best compromise between the number of zeros $\{0\}$ well classified on the left of the index idL (specificity) and the number of ones

$\{1\}$ well classified on the right of the index idL (sensitivity). This procedure is directly reflected on the output of the categorization process$\underline{\hat{t}}$, provided by the rule (6), resulting from the dichotomization of$\underline{d}$, using the threshold L, as shown in Fig. 4. Consequently, in an ideal classification process, all $\underline{t}$ values on the left of the index idL will be $\{0\}$, and all $\underline{t}$ values on the right of the index idL will be $\{1\}$, i.e., the vectors $\underline{t}$ and $\underline{\hat{t}}$ would present the same values. In a non-ideal situation, the estimation vector $\underline{\hat{t}}$ presents some differences with the target vector $\underline{t}$. Fig. 4 illustrates this process, where two values of zeros $\{0\}$, $\{t_4, t_6\}$, and one value of ones $\{1\}$, $\{t_{21}\}$ are incorrectly classified.

Using this approach, the maximization of the geometric mean (10) is achieved with the index idL that maximizes the product of the number of zeros $\{0\}$ well classified on the left of idL, and the number of ones $\{1\}$ well classified on the right of idL. Consequently, Eq. (10) is equivalent to (11).

$$\max_{0\leq L\leq 1} GM^2 = \max_{idL}\left\{csum\left(1-\underline{\hat{t}}\right)\circ\left(T_1 - csum\left(1-\underline{\hat{t}}\right)\right)\right\} \tag{11}$$

The operator $csum(\boldsymbol{a})$ computes the cumulative sum along the vector $\boldsymbol{a}$ of values equals to $\{1\}$, according to (12), where the operator (o) denotes a multiplication element by element. The constant T_1 identifies the total number of death patients, i.e., classified as $\{1\}$, easily computed as $sum\left(\underline{t}\right)$.

$$csuma_i = \sum_{k=1}^{i} a_k = 1 i = 1, \ldots, N \tag{12}$$

Therefore the vector $csum\left(1-\underline{\hat{t}}\right)\epsilon R^{N,1}$ computes the number of well classified zeros on the left of the index idL (specificity); the vector $\left(T_1 - csum\left(1-\underline{\hat{t}}\right)\right)\epsilon R^{N,1}$ computes the number of well classified ones on the right of the index idL (sensitivity). The optimal balance (geometric mean) is then obtained by maximizing the product of these two vectors, as defined by (11).

As result of this maximization process the idl index is obtained and, consequently, the threshold L. Considering the set of M rules (a rule for each risk factor), each rule j for each patient i, r_{ij}, can be optimally described as (13).

$$r_{ij} : \ if \ dx_{\mathrm{j}} \geq L_{ij} \ then \ \hat{t}_{ij} \quad j = 1, \ldots, \mathrm{M} \tag{13}$$

The variable dx_{ij} represents the normalized distance of the j^{th} risk factor for the patient i computed using (5), and L_j the threshold for rule j, determined by (11). The outcome $\widehat{t_{ij}}$ represents the estimated mortality risk for the patient i, using the rule j. The result provided by Eqs. (11) and (13) is of major relevance. In effect, by means of such formulation it is possible to obtain for each rule the optimal threshold that maximizes the GM, avoiding the implementation of an exhaustive search, and the necessary computational time to the GM determination at each iteration. Actually,

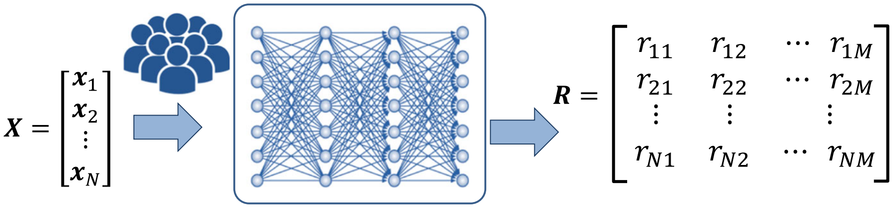

Fig. 5 Training a NN to estimate the correctness of the set of rules

using this formulation, only a sort operation involving a vector is required, together with the determination of the maximum of a vector that results from the element-wise multiplication of two vectors.

Finally, it should be stressed that the clinical significance of all rules and the respective thresholds have been validated and, in some cases, adjusted taking into account the recommendations of the clinical partner. Therefore, it is possible to guarantee that the baseline of the derived rules is in agreement with the expertize of clinicians.

3.3.2 Phase 2: Rules selection—Personalization issue

Training of predictive models Considering the set of baseline rules in the form of (4) or (13), the second step consists of training a ML model. The objective is the identification of the most proper rules for a given patient (personalization).

To this aim, the target of the ML model is composed of the correctness of the rules. This is a fundamental difference with respect to the standard approach (Fig. 1), where the mortality risk class is directly applied as the target of the classification process. It should be noted that in the training phase, the target output, $t_i = \{0, 1\}$, is known, as well as the output of each rule $r_i = \{0, 1\}$. As result, the correctness (*error*) of each rule can be easily assessed as a {yes/no} = {0,1} response. For example, for a 90 years old patient who survived ($t_i = 0$), and a rule defined as if AG $\geq$ 80 then $\widehat{t_i} = 1$, it is possible to conclude that this rule is not correct ($error = t_i - \widehat{t_i} = 0 - 1$).

Therefore, the training phase assumes as inputs the risk factors of the patients, $\boldsymbol{X} \epsilon R^{N,M}$, and the target output is defined as the correctness (yes or no) of the rules $\boldsymbol{R} \epsilon R^{N,M}$ (a set of M rules is assumed, one per each risk factor). In other words, the ML models do not learn how to classify the risk of a given patient but, instead, they are trained to determine if a baseline rule is correct or not, when applied to a specific patient. Figure 5 outlines this process where the variable $r_{ij} \epsilon \{0, 1\}$ defines the correctness of the rule j for the patient i.

As mentioned, this step is the key point of the present strategy. In to some extent, it simulates the reasoning of a physician: he does not apply all his expertise (described as rules) in the same way to all patients. Instead, he restricts his expertise (through the subset of rules) in order to perform a tailored diagnosis, considering the particular characteristics of each patient. Note this is completely different from the standard ML approach (Fig. 1), since the explainability of the rules is preserved. Moreover, only the rules which are more likely to be suitable to a patient are selected.

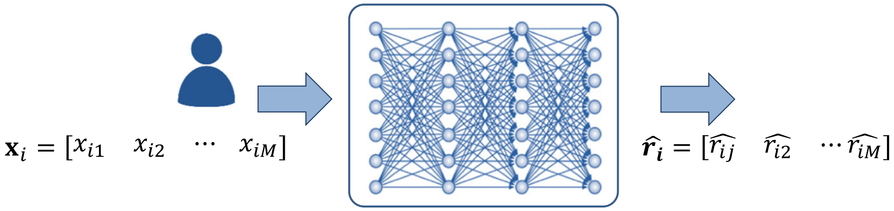

Fig. 6 Estimating the correctness of each rule for a specific patient

Table 1 Estimation of the rule output ($\hat{t}_{ij}$) and respective rule acceptance ($\hat{r}_{ij}$)

	Rule 1 $AG > 80$	Rule 2 $GD =$ male	Rule 3 $NC \geq 3$	Rule 4 *Male &NC* $= 2$
Target—t_i	0 (survival)			
Rule output—$\hat{t}_{ij}$	0	1	0	0
Rule acceptance, i.e., ML output—$\hat{r}_{ij}$	1	1	0	1

Assessment of rules acceptance for a new patient After being trained, and given the characteristics (risk factors) of a new patient, $\boldsymbol{x}_i \epsilon R^{1,M}$, the ML output, $\widehat{\boldsymbol{r}}_i \epsilon R^{1,M}$, provides an estimation of the correctness of each one the baseline rules. Therefore, the outcome $\widehat{r}_{ij}$, defines the acceptance (i.e., the predicted correctness) of the rule j for the new patient i, a binary variable $\{0,1\}$, as depicted in Fig. 6.

A value of $\widehat{r}_{ij}$=1 means that the corresponding rule output, Eq. (13), should be applied to assess the mortality risk of that patient (rule is accepted), as the rule is expect to be correct. In contrast, a value of $\widehat{r}_{ij}$=0 means that the corresponding rule output, Eq. (13), should not be used to assess the mortality risk of the current patient (rule is not accepted), as the rule is expect to be incorrect.

The outputs of the accepted rules, will be combined to estimate the mortality risk $\widehat{t}_i$ of a patient. To illustrate this procedure, let us consider the existence of four rules ($r_{ij}\ j = 1,.0.4$) to estimate the outcome of a patient i:

$$
\begin{aligned}
&\mathrm{r}_{\mathrm{i}1} : if\ \ AG \geq 80\ then\ \hat{t}_{i1} = 1\\
&\mathrm{r}_{\mathrm{i}2} : if\ \ GD = male\ then\ \hat{t}_{i2} = 1\\
&\mathrm{r}_{\mathrm{i}3} : if\ \ NC \geq 3\ then\ \hat{t}_{i3} = 1\\
&\mathrm{r}_{\mathrm{i}4} : if\ \ (GD = male\ AND\ NC = 2)\ then\ \hat{t}_{i4} = 1
\end{aligned}
$$

Considering a patient who survived ($t_i = 0$), characterized by AG = 70, GD = male, and NC = 1, Table 1 presents the rules output and the respective ideal acceptance for this illustrative case.

The output of each rule $\hat{t}_{ij}$ is directly evaluated using the characteristics (risk factors) of the patient (AG = 70, GD = male, and NC = 1) and the definition of each one of the four rules. The rule acceptance $\hat{r}_{ij}$, provided by the ML model (simulated values in the example), defines which rules should be used to estimate the patient' outcome, $\hat{t}_i = \{death(1), survival(0)\}$. In this demonstrative example, the ML recognizes that rules r_{i1}, r_{i2}, and r_{i4} are identified as correct, so these rules should be combined to estimate the value of $\hat{t}_i$. As result, if a majority voting scheme is used, the estimate patient's mortality risk will be $\tilde{y}_i = \{survival = 0\}$, obtained from the most common output of the accepted rules, $\tilde{y}_i =$ voting $(0, 1, -, 0\} = 0$.

3.3.3 Phase 3: Estimation of mortality risk—reliability issue

The third phase combines the accepted rules in the previous phase, $\hat{r}_{ij}$, to derive an estimation for the mortality risk of the patient. The classification scheme proposed here aims to enhance this assessment, by means of a mortality estimation and the reliability associated with that prediction.

Estimation of the mortality score The mortality score ($\tilde{y}_i$) proposed here is obtained for each patient i according to (14), using the subset of the T rules ($T \leq M$) that have been accepted, i.e., verifying $\hat{r}_{ij} = 1$.

$$\tilde{y}_i = \frac{1}{T} \sum_{j=1}^{T} \hat{t}_{ij} \hat{r}_{ij} \tag{14}$$

Basically, this score can be defined as the rate of accepted rules that suggest the patient's death, i.e., the ratio between the number of rules that have been accepted and suggest the death of the patient ($\hat{r}_{ij} = 1 \cap \hat{t}_{ij} = 1$) and T, the number of all rules that have been accepted ($\hat{r}_{ij} = 1$).

Calibration process Ideally, if the patient dies all the positive rules should be accepted and the negative ones should not be selected, and if the patient survives the opposite situation should occur. Therefore, the mortality score $\tilde{y}_i$ is expected to tend to $\{0\}$ in case of survival, or to tend to $\{1\}$ in case of death. In the proposed methodology, new labels have been introduced in the second phase (the acceptance of the rules), instead of mapping directly the features into the outcomes (mortality risk). Therefore, an additional calibration step is required to convert the estimated score $\tilde{y}_i$, given by (14), into an actual estimated risk ($\hat{y}_i$).

Calibration techniques, as post processing techniques, aim at improving the probability estimation or the error distribution of an existing model, thus contributing to enhance model's confidence (Bella et al. 2019). Depending on the task, different calibration techniques can be applied (Bella et al. 2019). A common solution consists of a logistic regression model, based on the estimated scores in the training dataset and their corresponding true label (death or survival), where two specific parameters β_0 and β_1 are computed. The logistic regression calibration model is applied here, Eq. (15), using the predicted score to obtain an estimation of the mortality risk (16).

$$\hat{y}_i = \frac{\exp(\beta_0 + \beta_1 \tilde{y}_i)}{1 + \exp(\beta_0 + \beta_1 \tilde{y}_i)} \tag{15}$$

Finally, a binary estimation for the mortality risk, i.e., $\hat{t}_i$ ={*survival*, *death*}, is directly obtained as (16) using (15).

$$\hat{t}_i = \left(\hat{y}_i \geq P\right) \tag{16}$$

The value of $\hat{y}_i$ represents the patient's risk of death estimation, a value in the range [0.0.1], and P is the threshold that binarizes the risk. This binary classification is the one used to compute specificity and sensitivity, and consequently the geometric mean. The threshold P can be obtained with several forms, sometimes depending on the goal of the model (for example, maximize specificity or sensitivity). For sake of simplicity, here we consider a threshold based on death rate. Indeed, if a given cohort has a rate of mortality of σ, then it is reasonable that a patient with a mortality estimation higher than σ is a high risk one, and a patient with a risk lower than σ is a low risk one.

Reliability index for the estimated mortality risk Along with the creation of a personalized and interpretable mortality prediction computation, we intended to extend the explainability of such predictions through the assessment of the confidence of each individual forecasting. While discrimination metrics like AUC and GM provide information about the expected accuracy of the created model for the population in general, they do not provide insights for each patient. Calibration metrics, namely the calibration curve that plots the estimates against the true occurrences, may be used to that purpose. However, they assess how likely the predictions are to be over or underestimated but the score is the same to all the predictions with the same value (in this study, the patients with the same mortality risk).

In this study we are then interested in the computation of a score which provides information about how reliable a particular prediction for a given patient is expected to be, i.e. how trustful it is. In this way, the methodology can provide the physicians not only a mortality forecasting but also the degree of confidence of that assessment for a given patient. Therefore, the physician gets additional information about the behaviour of the algorithm, i.e. if it is more prone to be correct or to be wrong.

If a given patient dies then it is expected that the rules that suggest his/her death (positive class $\{1\}$) are more accepted than the rules that suggest his/her survival (negative class $\{0\}$). In fact, in the ideal scenario, the positive rules would all be selected and the negative rules would all be rejected. Following the same reasoning, the opposite is expected if the patients survives. Thus, the higher the difference between the acceptation of positive and negative rules, the more is the confidence that the algorithm mortality output is correct. Therefore, the reliability measure of a particular mortality prediction, $\hat{w}_i$, for a patient i, was computed as the absolute value of the difference between the mean of the acceptation of positive rules {output $\hat{t}_i = 1$, death} and negative rules {output $\hat{t}_i = 0$, survival}, as (17), where p is the number of rules that suggest the patient's death and q is the number of rules that suggest the patient's survival.

$$\hat{w}_i = \left| \frac{1}{p} \sum_{j=1}^{p} \hat{r}_{ij} - \frac{1}{q} \sum_{j=1}^{q} \hat{r}_{ij} \right| \tag{17}$$

Table 2 Two examples of how to obtain and explain the reliability estimation

	Patient 1		Patient 2	
	Rule output $\hat{t}_{ij}$	ML output $\hat{r}_{1j}$	Rule output $\hat{t}_{ij}$	ML output $\hat{r}_{2j}$
Rule 1: AG > 80	1	0	1	0
Rule 2: GD = male	1	1	1	1
Rule 3: NC ≥ 3	0	1	1	1
Rule 4: Male &NC = 2	0	1	0	0
Mean acceptance of positive rules (death)	0.50		0.66	
Mean acceptance of negative rules (survival)	1		0	
Estimated reliability $\hat{w}_i$	0.50		0.66	

Table 2 presents an example of two patients, and the respective predicted correctness (acceptance given as the ML output) of the four arbitrary rules previously presented in Sect. 3.3.2). Let us consider that the patient 1 is characterized as $AG = 85$, $GD =$ male, and $NC = 1$ and the patient 2 is characterized as AG = 85, GD = male, and $NC = 3$. Additionally, let us consider as an example, the acceptance values ($\hat{r}_{ij}$) depicted in Table 2. Then, the combination of both rules outputs and their acceptance provides a reliability estimation. In opposition to the mortality score, the determination of this reliability index considers all rules.

4 Conclusion

The Fig. 7 summarizes the three main phases that characterize the current ML based approach.

In the first phase, a baseline set of explainable rules is specified, Eq. (13). In the current work a set of normalized rules was proposed (one rule per risk factor), although any type of interpretable rules could be easily incorporated in the presented strategy.

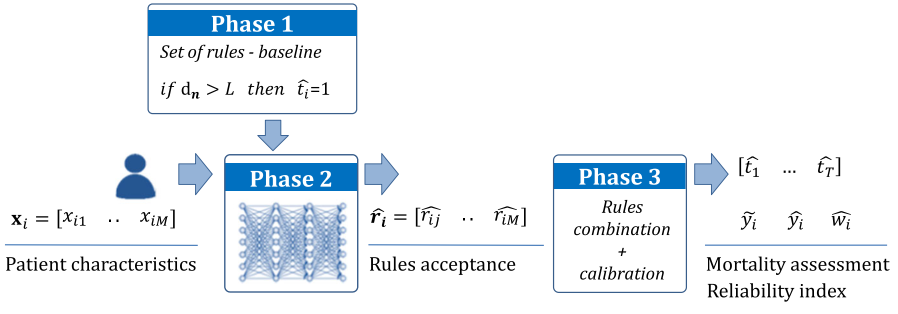

Fig. 7 Overview of the global scheme

This approach reflects a typical clinical prognosis: a patient is at risk of death if his/her characteristics (risk factors) are more similar (closer) to the "virtual patient" representative of the dead patients, than to the "virtual patient" representative of the survival patients. Moreover, it is possible to guarantee that a given rule presents an optimal performance (considering GM) without impairing its interpretability.

In the second phase, a ML-based scheme is proposed to estimate the acceptance of the set of interpretable baseline rules (derived in phase 1), when applied to a particular patient. First of all, this strategy directly addresses the personalization issue. In effect, by means of this scheme, only the rules that are more appropriated for a patient are used (accepted rules), thus avoiding the use of all rules (principle one fits all). Moreover, the explainability of the baseline rules is fully preserved, since the approach does not modify the rules, as it only selects a reduced set of them.

Finally, in the third phase, based on the combination of the output of the rules and the respective acceptance, it is possible to estimate a global value for the mortality risk, $\hat{y}_i$, together with a reliability index,$\hat{w}_i$, for this outcome.

5 Results

5.1 Dataset and baseline set of rules

5.1.1 Dataset

The proposed methodology was applied to a real problem related with the risk prediction of the 30-days all-cause mortality after an acute coronary syndrome (ACS) event. The data set is composed of N = 1111 patients resulted from two different cohorts, both from Portuguese hospitals, the Santa Cruz Hospital, Lisbon, and the Hospital de Santo André, Leiria. As result, the data set consists of a set of features (inputs) $\boldsymbol{X} \in R^{N,M}$ (risk factors), being the target (output) $\boldsymbol{t} \in R^{N,1}$, the mortality risk (30-days all cause death). The features includes several types of information, such as demographics, baseline characteristics, previous medication and the presence of comorbidities. The overall information about the variables considered in this study is presented in Table 3.

In this study the 30-days all-cause death rate is 4.95% (55 deaths in 1111 patients). For the binary variables (diabetes, smoking, hypertension, stroke/TIA, myocardial infarction, unstable angina, PTCA, CABG, ST-segment deviation, elevation of cardiac markers, all types of medication, and reperfusion) the shown values are related to the positive class. For the continuous variables, the reported values are related with the median and interquartile range. For the categorical variables the values describe the frequencies and the respective percentages.

5.1.2 Baseline set of rules

From the original set of risk factors, we have selected as risk factors the variables that present simultaneously a low p-value ($p < 0.01$) and are used in at least one of the three risk score models most applied in the daily clinical practice (GRACE, TIMI,

Table 3 General information of the patients used in the analysis of this study

	Total	Survival	Death	*p* value
No. of patients	1111	1056	55	
Diagnosis				
UA	247 (22.2%)	244 (23.1%)	3 (5.5%)	< 0.001
NSTEMI	452 (40.7%)	433 (41.0%)	19 (34.5%)	
STEMI	402 (36.2%)	369 (34.9%)	33 (60.0%)	
Demographics				
Gender				
Female	260 (23.4%)	240 (22.7%)	20 (36.4%)	0.020
Male	851 (76.6%)	816 (77.3%)	35 (63.6%)	
Age	64 (54–73)	64 (54–73)	72 (63–81)	< 0.001
Risk factors/comorbidities				
Diabetes	280 (25%)	261 (24.7%)	19 (34.5%)	0.106
Smoking—current smoker	288 (25.9%)	279 (26.4%)	9 (16.4%)	0.094
Smoking – former smoker	268 (24.1%)	261 (24.7%)	7 (12.7%)	0.038
Hypertension	695 (62.6%)	658 (62.3%)	37 (67.3%)	0.485
PAD	76 (6.8%)	70 (6.6%)	6 (10.9%)	0.207
Clinical antecedents				
Stroke and/or TIA	85 (7.7%)	71 (6.7%)	14 (25.5%)	< 0.001
Myocardial infarction	327 (29.4%)	314 (29.7%)	13 (23.6%)	0.367
Unstable angina	163 (14.7%)	159 (15.1%)	4 (7.3%)	0.120
PTCA	233 (21.0%)	227 (21.5%)	6 (10.9%)	0.067
CABG	128 (11.5%)	126 (11.9%)	2 (3.6%)	0.064
Characteristics presentation				
SBP (mmHg)	140 (121–160)	140 (121–160)	130(107–150)	< 0.001
DBP (mmHg)	80 (70–90)	80 (70–90)	70 (63–82)	< 0.001
Heart rate (bpm)	75 (64–88)	75 (64–87)	90 (76–25)	< 0.001
Killip class				
I	915 (82.4%)	888 (84.1%)	27 (49.1%)	< 0.001
II	103 (9.3%)	92 (8.7%)	11 (20.0%)	
III	65 (5.9%)	54 (5.1%)	11 (20.0%)	
IV	10 (1.0%)	6 (0.6%)	4 (7.3%)	
ST deviation	745 (67.1%)	700 (66.3%)	45 (81.8%)	0.013
LEVF				
> 50%	677 (60.9%)	664 (62.9%)	13 (23.6%)	< 0.001
[30,50]%	293 (26.4%)	278 (26.3%)	15 (27.3%)	
< 30%	80 (7.2%)	70 (6.6%)	10 (18.2%)	

Table 3 (continued)

	Total	Survival	Death	*p* value
Laboratory results				
Elevat. markers*a*	614 (55.3%)	572 (54.2%)	42 (76.4%)	0.001
Creatinine(mg/dL)	1.0 (0.9–1.2)	1.0 (0.9–1.2)	1.1 (1.0–1.5)	0.066
Glucose (mg/dL)	126 (101–169)	125 (101–167)	144 (123–209)	< 0.001
Hemoglobin (g/dL)	14.0(12.8–15.0)	14.0(12.9–15.0	12.8(11.8–14.4)	0.001
Medication pre-admission				
Antiplatelet drugs	526 (47.3%)	507 (48.0%)	19 (34.5%)	0.094
Beta-blockers	292 (26.3%)	279 (26.4%)	13 (23.6%)	0.715
Calcium channel blockers	245 (22.1%)	234 (22.2%)	11 (20.0%)	0.767
Statin	254 (22.9%)	242 (22.9%)	12 (21.8%)	0.918
Antihypertensive drugs	325 (29.3%)	307 (29.1%)	18 (32.7%)	0.485
Reperfusion	621 (55.9%)	596 (56.4%)	25 (45.5%)	0.110

UA: unstable angina, NSTEMI: non-ST-elevation myocardial infarction, STEMI: ST-elevation myocardial infarction, PAD: peripheral artery disease, TIA: transient ischemic attack, PTCA: percutaneous transluminal coronary angioplasty, CABG: coronary artery bypass grafting, SBP: systolic blood pressure, DBP: diastolic blood pressure, LVEF: left ventricular ejection fraction

[a]The elevation of cardiac markers is related to the elevation above critical thresholds of cardiac injury biomarkers, such as troponin, myoglobin and creatinine kinase (CK)-MB, accordingly to clinical guidelines

PURSUIT) (Gonçalves et al. 2005). In this way, we assure that we are dealing with variables that already prove its clinical importance in large worldwide datasets, and that are also significant in the cohort that is being analysed. Thus, six risk factors are considered: 1) Elevated ST segment diagnosis (STEMI), 2) age (AG), 3) systolic blood pressure (SBP), 4) heart rate (HR), 5) Killip class (KC), and 6) previous stroke/TIA (PS).

As referred in Sect. 3.3.1 the baseline set of rules considered here is based on the individual risk factors and the respective thresholds. In particular, the specific values of the thresholds have been computed using an optimization process (geometric mean maximization), afterwards adjusted in accordance with the recommendations of clinical experts. As result, the baseline is composed of a set of six simple rules, given by (18):

$$
\begin{aligned}
& r_1 : if\ \ STEMI = 1\ then\ \hat{t} = 1 \\
& r_2 : if\ \ AG \geq 68 \text{ years } then\ \hat{t} = 1 \\
& r_3 : if\ \ SBP \geq 135 \text{ mmHg } \ then\ \hat{t} = 1 \\
& r_4 : if\ \ HR \geq 84 \ \text{ bpm } then\ \hat{t} = 1 \\
& r_5 : if\ \ KC \geq 2\ then\ \hat{t} = 1 \\
& r_6 : if\ \ PS = 1\ then\ \hat{t} = 1
\end{aligned} \tag{18}
$$

5.2 Experimental results

5.2.1 Validation strategy

The methodology presented in this work was validated through a Monte Carlo stratified cross-validation (MCCV) with 1000 runs, where in each run 80% of the dataset was used to train and 20% to test. In order to assess the performance of the proposed methodology, the geometric mean (GM) and area under the receiver operating characteristic curve (AUC) were considered. These metrics were also computed for two other comparative models being the risk prediction performance also assessed using calibration curves (Sect. 4.2.2). The goodness of reliability estimation was analysed comparing it with the misclassifications rate (Sect. 4.2.3). The interpretability is discussed in Sect. 4.2.4 being, in Sect. 4.3, discussed the advantages of the proposed approach.

The imbalanced nature of this problem suggests that the rules more likely to be correct for the patients who survive would be accepted more often, even for the positive patients. To address this issue an under-sampling strategy was applied (ratio of 1.5/1 negative-to-positive samples). The imputation of values for missing data was based on a 10-nearest-neighbour imputation method.

Models to be compared The performance of the proposed approach was compared with two other methods: a clinical risk score (GRACE), and a standard ML model, namely an artificial neural network (ANN).

Proposed approach As explained, in the current strategy an ANN model is applied to estimate the acceptance of each rule for a given patient, enabling to select and use only the rules that are more appropriated to the diagnosis.

GRACE This clinical model was derived from a large cohort of patients with ACS admission (Granger 2345), and it is widely used in clinical practise. Furthermore, it showed to be the one that best estimates the 30-days mortality in a Portuguese cohort (Gonçalves et al. 2005). As any risk score model, the GRACE is based on the conversion of the logistic regression parameters to a system of points. Due its simplicity and very interpretable approach, those models are still a rule of thumb in clinical domain. In this study, the GRACE model was applied to the complete data (patients with missing values in the variables used by the model were discarded): 1013 patients, with an associated mortality rate of 4.54%.

Artificial neural network: For implementing a standard machine learning approach, illustrated in Fig. 1, a standard ANN (with a simple 3-layers architecture and sigmoidal activation functions) was considered. In this case, the model has the original risk factors values as inputs (X) with the binary label corresponding to death or survival after 30-days from admission as output (t).

5.2.2 Predictive performance results

The results of the 30-days mortality risk estimation using those three methodologies were analysed in terms of discrimination and calibration results. The AUC and GM results, over the 1000 runs of Monte-Carlo cross-validation, are presented in Table 4,

Table 4 Mortality risk estimation: proposed approach, ANN and GRACE models—(mean ± confidence interval)

	Proposed approach		ANN		GRACE	
	Train	Test	Train	Test	Train	Test
AUC	82.8 ± 0.1	79.3 ± 0.3	81.7 ± 0.3	78.4 ± 0.4	83[a]	78
GM	78.2 ± 0.1	73.4 ± 0.3	76.1 ± 0.3	69.7 ± 0.5	NA[a]	47[b]

[a]For the GRACE model, the training performance metrics were obtained from the original GRACE study and the geometric mean was not assessed (NA)

[b]The GRACE model categorizes the patients into three groups: low, intermediate and high risk. To compute the GM, the intermediate-risk group was joined to the low-risk

and the associated boxplots of testing data are presented in Figs. 8 and 9, respectively for the AUC and for the GM metrics.

Considering the obtained results, the proposed procedure showed to perform very well in estimating the mortality risk after an ACS event in terms of discrimination metrics (Table 4, and Figs. 8 and 9) and calibration assessment (Fig. 10). The latter shows how the mortality rate varies with the predicted mortality risk in the validation group. As can be observed, the proposed methodology performs slightly better than the standard ANN, in terms of AUC, GM and calibration. Furthermore, the proposed approach seems to be much more robust than the ANN as the latter presents a very significant number of outliers with a very lower performance, as shown in Figs. 8 and 9. The proposed approach also presents a considerably better performance than the GRACE model. While GRACE performance may be negatively affected not only because it was derived from a more heterogeneous population but also uses 3 categories instead of only two, GRACE model shows a very poor geometric mean and a high underfitting in the calibration assessment. Such considerations seem to suggest that

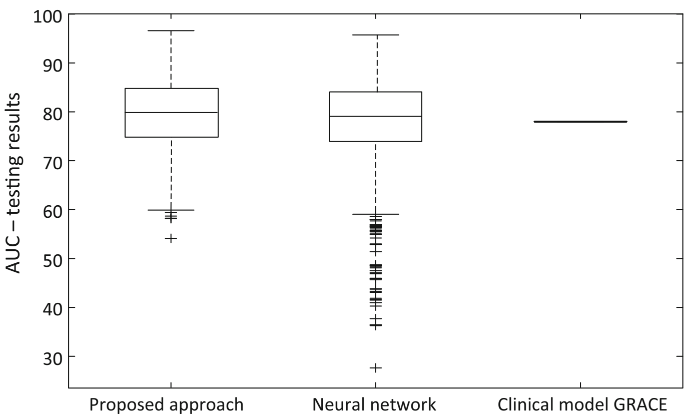

Fig. 8 AUC performance for the proposed approach, ANN and GRACE

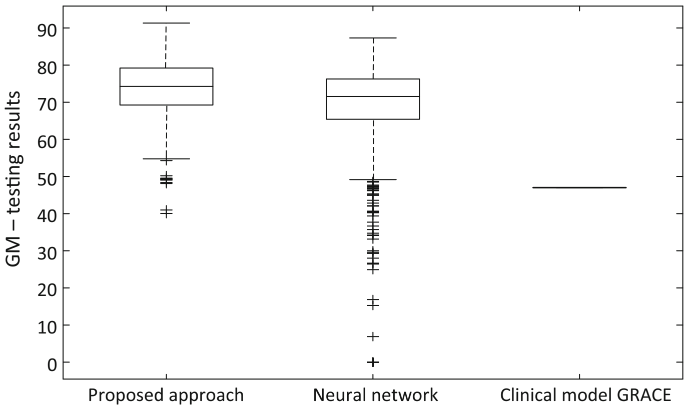

Fig. 9 GM performance for proposed approach, ANN and GRACE

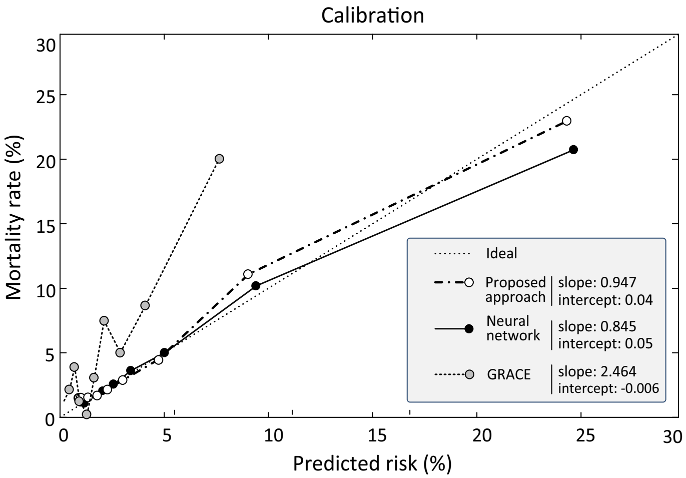

Fig. 10 Calibration curve in the testing dataset. For the proposed approach and ANN each point represents the mean and 95% confidence intervals of the means of the deciles of predicted risk values, over the 1000 runs of MCCV. For GRACE each point represents the mean of the deciles of the predicted risk values, for the entire dataset

even it is widely used in the Portuguese hospitals, a more precise model may be required.

5.2.3 Reliability estimation

The assessment of how misclassification rate varies depending on the computed reliability estimates, is represented in Fig. 11 for the testing group. The misclassification rate corresponds to the percentage of patients who were incorrectly classified: patients who died and were predicted to survive, or patients who survived and were predicted

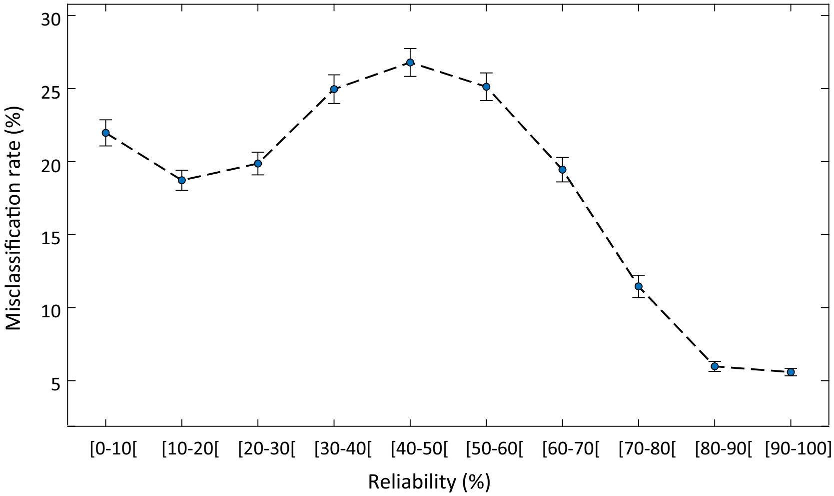

Fig. 11 Variation of misclassifications rate according to the reliability estimates, for the testing dataset. The reliability values were divided into subgroups of 10% for easier interpretation. Each point represents the mean and 95% confidence intervals over the 1000 runs of MCCV

to die. This binary classification (into low risk or high risk group) was performed as explained before.

The proposed solution shows to efficiently estimates the reliability of the individual predictions. It is possible to observe that the physicians can trust the algorithm when the reliability associated to the predictions is very high: for a reliability index of 80% or higher, only approximately 5% of the cases the risk classification produced by the algorithm is not correct. On the other hand, for a reliability index lower than 70%, the algorithm is wrong in approximately 20–25% of the cases. This is very significant as the algorithm itself is able to inform the physician about the reliability of its own predictions. While for the first scenario (reliability score > 80%), the algorithm is very likely to be correct, in the second scenario (reliability score < 70%), a more cautious acceptation of the algorithm output is required and extra expertise clinical knowledge may be necessary to assess the forecasting of the patient condition.

An important note is that such reliability estimation is not commonly addressed in machine learning studies. Indeed, both clinical (GRACE) and ML (ANN) models do not naturally compute this score: an auxiliary procedure is required. Even some external procedures can be computed, there is some lack of its research in ML area. However, as shown through this practical scenario, its importance in critical decision-making, namely in the clinical context, is undeniable.

5.2.4 Interpretability evaluation

As stated, one of the main motivations of this work is to propose a strategy where the model is composed of a set of interpretable rules, able to offer high explainability to the physicians. To this aim a set of baseline rules was derived in the form of (13), by means

Table 5 Outputs suggested by each one of the created rules

Rules		
Diagnosis	UA/NSTEMI: survival	STEMI: death
Age	$<$ 68 years: survival	$\geq$ 68 years: death
SBP	$<$ 135 mmHg: death	$\geq$ 135 mmHg: survival
Heart rate	$<$ 84 bpm: survival	$\geq$ 84 bpm: death
Killip class	I: survival	II to IV: death
Previous stroke/TIA	No: survival	Yes: death

The levels of the categorical variables (diagnosis, Killip class and previous stroke/TIA) have an ordinal nature from a milder to a more severe condition. This allows such variables to be used similarly to the continuous variables

of the “virtual patient” concept, based on the centroids of each cluster/class (death and survival). If a patient is closer to one class it is reasonable to assume him/her as part of it. It should be noted that this principle (case based reasoning) is in accordance with current practice followed by physicians, thus directly interpretable and accepted by them.

In particular, six simple rules have been derived based on individual risk factors. For each rule a threshold L (enabling to discriminate the patient’s class) was obtained in terms of its absolute value, which originates the rules summarize in (18), and described in Table 5. Those rules are in accordance with what is considered in the state-of-art clinical risk models, as they are easy to interpret as well as they have clinical meaning, once that the thresholds have been clinically confirmed/adjusted.

In effect, for each patient the model consists of a subset of the baseline set of rules (Table 5) that are selected (and combined) taking into account the particular characteristics of each patient. Since each individual rule of the baseline is interpretable any subset is, unquestionably, also interpretable.

5.3 Discussion

As mentioned, the main goal of this approach is to address three of the fundamental principles of explainable AI (interpretability, personalization, and reliability), without impairing the performance of the model’s prediction.

Interpretability The clinical knowledge integrated into the proposed approach guarantees a strong interpretability of the final model. In effect, by means of the use of dichotomic rules, Eq. (4), it is possible to clearly describe, in a very explainable form, how the outcome is obtained. The particular rules used in this study, obtained in collaboration with the clinical team, are presented in (18) and described in Table 5.

Personalization The current strategy enables to estimate the acceptance of the baseline rules, when applied to a particular patient. As result, the strategy directly addresses the personalization issue: only the rules that are expected to be correct for a given patient are applied in the estimation of the mortality risk, thus avoiding the use of all rules (principle one fits all).

Reliability The computation of confidence levels introduces the ability to predict the reliability of each particular estimation, which completes the use of calibration curves, Fig. 10. The results reveal that there is a significant potential to properly evaluate such reliability for each prediction, estimating when the algorithm is likely to produce a correct output or when it is more likely to be misleading, as depicted in Fig. 11.

Performance The integration of a ML model, capable to select only the rules that are more appropriated to be combined in the final diagnosis (Fig. 6), enables to achieve performances similar to the ones obtained with the application of ML models in a standard structure, Fig. 2. Moreover, the computation of an optimal threshold, through an optimization process, Eq. (9), can contribute to this performance.

Finally, Table 6 illustrates how this methodology could be applied in the clinical practice. This scenario represents the data of a real patient admitted with acute coronary syndrome. The rule output is obtained from the observed features, considering the thresholds presented in Table 5. The rule acceptance is then computed through the ML models, predicting when each rule is expected to be correct or not. For this patient, five rules are accepted (all except SBP), i.e. they are anticipated to provide a true output. Therefore, only those five rules are used to forecast the patient mortality risk. From those, four rules suggest his/her survival. Take this into consideration, the mortality risk is computed, Eq. (15). As that risk is lower than the rate of patients who died (4.95%), the patient is considered to be a low-risk one. Furthermore, the reliability estimation is computed based on the total number of rules that suggested the patient's death (two rules, one selected) as well as on the total of rules that suggested his/her survival (four rules, all selected).

Table 6 Application of the proposed approach to a real patient admitted with ACS

Risk factor/rule	Observed value	Rule output	Rule acceptance
Diagnosis	UA	Survival	Yes
Age (years)	71	Death	Yes
SBP (mmHg)	130	Death	No
Heart rate (bpm)	75	Survival	Yes
Killip class	I	Survival	Yes
Antecedent of stroke or AIT	No	Survival	Yes
Predicted mortality risk: 1.7%			
Categorization: low-risk group – {survival}			
Predicted reliability: 50%			

Summing up, this is a generic and powerful approach towards explainable AI, able to provide a tool to assist the decision making process, and to definitely increase the trust in this type of tools. For explanation purposes, a clinical case study was applied throughout this manuscript. However, it is very important to underline, that this is a global approach that can be easily applied to several classification problems in a broad set of different contexts.

6 Conclusions

The development of methodologies capable to introduce transparency and reliability concerns, with the capacity to better explain a model, and justify its outcomes is of decisive importance. Explainable AI has emerged recently as a research field, aiming at addressing these issues.

Towards explainable AI, the current work proposes an innovative machine learning based approach, implementing a hybrid scheme, capable to combine knowledge-driven and data-driven techniques. Through this approach three of the main explainable AI issues were addressed (interpretability, personalization, and reliability), while keeping the superior performance of ML methods. The existence of appropriate answers for those issues may be peremptory as a basis for auditability processes, and for its acceptance to support critical tasks, namely in critical decision-making scenarios, e.g. some clinical contexts.

The developed methodology is generic, able to be applied in other contexts, where "black boxed" ML approaches present limitations. This study can be considered as a first exploratory step, being the proposed ideas validated in the field of cardiovascular risk assessment, namely in the prediction of patient's mortality risk (30 days) after an acute coronary syndromes event. For future research, additional studies using other datasets will be carried out in order to further validate the effectiveness of the proposed strategy. In parallel, other techniques for improving the prediction and diagnosis performance will be investigated, namely the creation of more accurate rules that will be based on the combination of different risk factors. Finally, one of the main directions of work will be oriented to the quantification of the model interpretability, as well as, its comparison with other models.

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References

Adadi A, Berrada M (2018) Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). IEEE Access 6:52138–52160

Adebayo J, Gilmer J, Muelly M, Goodfellow I, Hardt M, Kim B (2018) Sanity checks for saliency maps. Adv Neural Inf Process Syst 568:9505–9515

Ahmad M, Eckert C, Teredesai A, McKelvey G (2018) Interpretable machine learning in healthcare. IEEE Intell Inf Bull 19(1):596

Arrieta B (2019) Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion. https://doi.org/10.1016/j.inffus.2019.12.012,2019
Barocas S, Hardt M, Narayanan A (2001) Fairness and machine learning limitations and opportunities. In: Smatinc scholar proceedings, ID: 113402716
Bella A, Ferri C, Orallo J, Quintana M (2019) Calibration of Machine learning models. In: Handbook of research on machine learning applications. IGI Global
Bradley A (1997) The use of the area under the ROC curve in the evaluation of machine learning algorithms. Pattern Recogn 30:1145–1159
Breiman L, Friedman J, Olshen R, Stone C (1984) Classification and regression trees. Wadsworth Inc
Burkart N, Huber M (2021) A Survey on the explainability of supervised machine learning. J Artif Intell Res 70:245–317
Burrell J (2016) How the machine thinks: Understanding opacity in machine learning algorithms. Big Data Soc 3(1):568
Carrington A, Manuel D, Fieguth P, Ramsay T, Osmani V, Wernly B, Bennett C, Hawken S, McInnes M, Magwood O, Sheikh Y, Holzinger A (2021) Deep ROC analysis and AUC as balanced average accuracy to improve model selection, understanding and interpretation. In: IEEE transactions on pattern analysis and machine intelligence. arXiv:2103.11357
Caruana R, Lou Y, Gehrke J, Koch P, Sturm M, Elhadad N (2015) Intelligible models for healthcare: predicting pneumonia risk and hospital 30-day readmission. In: Proceedings 21th SIGKDD International conference on knowledge discovery and data mining, pp 1721–1730
Carvalho D, Pereira E, Cardoso J (2019) Machine learning interpretability: a survey on methods and metrics. Electronics 8:832. https://doi.org/10.3390/electronics8080832
Casalicchio G, Molnar C, Bischl B (2018) Visualizing the feature importance for black box models. In: Joint European conference on machine learning and knowledge discovery in databases. Springer, pp 655–670 (2018)
Cohen W (1995) Fast effective rule induction. Semantic Scholar. https://doi.org/10.1016/b978-1-55860-377-6.50023
Dandl S, Molnar C, Binder M, Bischl B (2020) Multi-objective counterfactual explanations. arXiv preprint arXiv:2004.11165
Doshi-Velez F, Towards K (2017) A rigorous science of interpretable machine learning. arXiv:1702.08608
Dubey R, Zhou J, Wang Y, Thompson PM, Ye J (2014) Analysis of sampling techniques for imbalanced data: An n= 648 adni study. Neuroimage 87:220–241
European Commission (2020) White Paper On Artificial Intelligence—a European approach to excellence and trust. https://ec.europa.eu/info/sites/info/files/commission-white-paper-artificial-intelligence-feb2020en.pdf
Freitas A (2014) Comprehensible classification models: a position paper. ACM SIGKDD Explor Newsl 15(1):1–10
Frosst N, Hinton G (2017) Distilling a neural network into a soft decision tree. arXiv preprint arXiv:1711.09784
Gavish B, Ben-Dov IZ, Bursztyn M (2008) Linear relationship between systolic and diastolic blood pressure monitored over 24 h: assessment and correlates. J Hypertens 26:199
Gonçalves P, Ferreira J, Aguiar C, Seabra-Gomes R (2005) TIMI, PURSUIT, and GRACE risk scores: sustained prognostic value and interaction with revascularization in NSTE-ACS. Eur Heart J 26:865
Granger C (2003) Predictors of hospital mortality in the global registry of acute coronary events. Arch Intern Med 163:2345
Guidotti R, Monreale A, Ruggieri S, Turini F, Giannotti F, Pedreschi D (2018) A survey of methods for explaining black box models. ACM Comput Surv
Hajian S, Bonchi F, Castillo C (2016) Algorithmic bias: From discrimination discovery to fairness-aware data mining. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. ACM, pp 2125–2126
Hajian-Tilaki K (2013) Receiver operating characteristic (ROC) curve analysis for medical diagnostic test evaluation. Caspian J Intern Med Spring 4(2):627–635
Hastie T, Friedman J, Tibshirani R (2001) The elements of statistical learning, volume 1. Springer series in statistics Springer, Berlin
Holmes G, Hall M, Prank E (1999) Generating rule sets from model trees. In: Australasian joint conference on artificial intelligence. Springer, pp 1–12

Holzinger A, Langs G, Denk H, Zatloukal K, Muller H (2019) Causability and explainability of artificial intelligence in medicine. Wiley Interdiscip Rev Data Min Knowl Discov 9(4):e1312. https://doi.org/10.1002/widm.1312
Holzinger A, Carrington A, Muller H (2020) Measuring the quality of explanations: the system causability scale (SCS). KI Künstliche Intell 34:193–198. https://doi.org/10.1007/s13218-020-00636-z
Holzinger A, Carrington A, Muller H (2021) Measuring the quality of explanations: the system causability scale (SCS). Inf Fusion 71:28–37
Holzinger A, Malle B, Saranti A, Pfeifer B (2021) Towards multi-modal causability with Graph Neural Networks enabling information fusion for explainable AI. Inf Fusion 71:28–37
Hornik K, Buchta C, Zeileis A (2009) Open-source machine learning: R meets Weka. Comput Stat 24(2):225–232. https://doi.org/10.1007/s00180-008-0119-7
Krishnan S, Wu E (2017) Palm: machine learning explanations for iterative debugging. In: Proceedings of the 2nd workshop on human-in-the-loop data analytics, pp. 1–6
Linardatos P, Papastefanopoulos V, Kotsiantis S (2021) Explainable AI: a review of machine learning interpretability methods. Entropy 23:18. https://doi.org/10.3390/e23010018
Lipton Z (2018) The mythos of model interpretability. Queue 16(3):31–57
Luque A, Carrasco A, Martína A, Heras A (2019) The impact of class imbalance in classification performance metrics based on the binary confusion matrix. Pattern Recognit 91:216–231
Margot V (2020) A rigorous method to compare interpretability. hal-02530389v5
Markus A, Kors J, Rijnbeek P (2020) The role of explainability in creating trustworthy artificial intelligence for health care: a comprehensive survey of the terminology, design choices, and evaluation strategies. J Biomed Inform. https://doi.org/10.1016/j.jbi.2020.103655
Mehrabi N, Morstatter F, Saxena N, Lerman K, Galstyan A (2019) A survey on bias and fairness in machine learning. arXiv:1908.09635 [cs]
Molnar C, Casalicchio G, Bischl B (2019) Quantifying model complexity via functional decomposition for better post-hoc interpretability. In: Joint European conference on machine learning and knowledge discovery in databases. Springer, pp 193–204
Molnar C. Interpretable machine learning. A guide for making black box models explainable. https://christophm.github.io/interpretable-ml-book/. ISBN 978-0-244-76852-2
Mothilal R, Sharma A, Tan C (2020) Explaining machine learning classifiers through diverse counterfactual explanations. In: Proceedings of the 2020 conference on fairness, accountability, and transparency, pp. 607–617
Murdoch W, Singh C, Kumbier K, Abbasi-Asl R, Yu B (2019) Definitions, methods, and applications in interpretable machine learning. PNAS J 11(44):22071–22080
O'Sullivan S (2019) Legal, regulatory, and ethical frameworks for development of standards in artificial intelligence (AI) and autonomous robotic surgery. Int J Med Robot Comput Assist Surg 15:1–12. https://doi.org/10.1002/rcs
Parliament and Council of the European Union. General data protection regulation. https://eur-lex.europa.eu/eli/reg/2016/679/oj
Philipp M, Rusch T, Hornik K, Strobl C (2018) Measuring the stability of results from supervised statistical learning. J Comput Graph Stat 27(4):685–700
Puri N, Gupta P, Agarwal P, Verma S, Krishnamurthy B (2017) Magix: model agnostic globally interpretable explanations. arXiv preprint arXiv:1706.07160
Qayyum A, Qadir J, Bilal M, Fuqaha A (2020) Secure and robust machine learning for healthcare: a survey. IEEE Rev Biomed Eng
Quinlan J (1986) Induction of decision trees. Mach Learn 1:81–106
Quinlan J (1992) C4.5 programs for machine learning. Morgan Kaufmann
Nassih R, Berrado A (2020) State of the art of Fairness, Interpretability and Explainability. In: 13th international conference on intelligent systems: theories and applications (SITA'20). September 23–24, 2020, Rabat, Morocco. ACM, New York, NY, USA
Ribeiro M, Singh S, Guestrin C (2016) Why should i trust you?: Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. ACM, pp 1135–1144
Ribeiro M, Singh S, Guestrin C (2018) Anchors: high-precision model-agnostic explanations. In: Thirty-second AAAI conference on artificial intelligence
Rudin C, Chen C, Chen Z, Huang H, Semenova L, Zhong C (2022) Interpretable machine learning: fundamental principles and 10 grand challenges. Stat Surv. arXiv:2103.1125

Schneeberger D, Stoger K, Holzinger A (2020) The European Legal Framework for Medical AI"; Springer Nature Switzerland AG 2020. In: Holzinger A et al (eds). https://doi.org/10.1007/978-3-030-57321-8_12
Selvaraju R, Cogswell M, Das A, Vedantam R, Parikh D, Batra D (2019) Grad-CAM: visual explanations from deep networks. Int J Comput vis. https://doi.org/10.1007/s11263-019-01228-7
Stanford University (2016) Artificial intelligence: trends and predictions for 2030. https://www.qulix.com/wp-content/uploads/2017/04/Artificial-intelligence-Trends-and-Predictions-for-2030.pdf
Strobl C, Boulesteix A, Kneib T, Augustin T, Zeileis A (2008) Conditional variable importance for random forests. BMC Bioinform 9(1):307. https://doi.org/10.1186/1471-2105-9-307
Strumbelj E, Kononenko I (2014) Explaining prediction models and individual predictions with feature contributions. Knowl Inf Syst 41(3):647–665
Tjoa E, Guan C (2015) A survey on explainable artificial intelligence: towards medical XAI. J Latex Class Files 14(8):564
Ustun B, Rudin C (2016) Supersparse linear integer models for optimized medical scoring systems. Mach Learn 102(3):349–391
Wang T, Rudin C, Velez F, Liu Y, Klampfl E, MacNeille P (2017) A Bayesian framework for learning rule sets for interpretable classification. J Mach Learn Res 18:256
Ying X (2019) An overview of overfitting and its solutions. IOP Conf Ser J Phys Conf Ser 1168-022022
Zhou J, Gandomi A, Chen F, Holzinger A (2021) Evaluating the quality of machine learning explanations: a survey on methods and metrics. Electronics 10:593. https://doi.org/10.3390/electronics10050593
Zhou Q, Liao F, Mou C, Wang P (2018) Measuring interpretability for different types of machine learning models. In: Paciffc-Asia conference on knowledge discovery and data mining, pp 295–308